structure, which is a CBOR array with the following elements:

- An OSCORE version number, currently always set to 1
- The COSE algorithm identifier
- The Sender ID
- The Partial IV
- An options string. This is used to encode CoAP headers but is blank in this example.

The COSE structure is then encrypted with the sender key.

> **DEFINITION** COSE allows messages to have ***external additional authenticated data***, which is included in the message authentication code (MAC) calculation but not sent as part of the message itself. The recipient must be able to independently recreate this external data otherwise decryption will fail.

Listing 13.12 Encrypting the plaintext

```
long sequenceNumber = 20L;
byte[] nonce = nonce(13, sequenceNumber, senderId,
commonIv);    #A
byte[] partialIv = new byte[] { (byte) sequenceNumber };
#A

var message = new Encrypt0Message();
message.addAttribute(HeaderKeys.Algorithm,    #B
        algorithm.AsCBOR(), Attribute.DO_NOT_SEND);    #B
message.addAttribute(HeaderKeys.IV,    #B
        nonce, Attribute.DO_NOT_SEND);    #B
message.addAttribute(HeaderKeys.PARTIAL_IV,    #C
        partialIv, Attribute.UNPROTECTED);    #C
message.addAttribute(HeaderKeys.KID,    #C
```